Priya Sadhu

Machine Learning Engineer

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Summary

Machine Learning Engineer with expertise in Python, deep learning, predictive analytics, and data analysis. Experienced in developing AI models for classification and anomaly detection using structured datasets. Passionate about building scalable, explainable, and production-ready AI solutions.

Skills

Programming & Querying: SQL, Python
Machine Learning & Deep Learning: Model Development, Feature Engineering, Model Evaluation and Validation
Web Development: HTML, CSS, React.js, FastAPI, Node.js
Libraries & Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib

Experience

AIML Intern (Artificial Intelligence & Machine Learning) **July 2025 – August 2025**
CTTC, Bhubaneswar *Tools Used: Python, NumPy, Pandas, OpenCV*

- **Engineered image preprocessing workflows** for computer vision applications using resizing, normalization, and grayscale conversion to enhance model performance and detection consistency.
- **Implemented a real-time face detection system** using OpenCV Haar Cascade classifiers, enabling robust face recognition across varying lighting conditions and complex backgrounds.
- **Built handwritten digit recognition models** using structured datasets, applying data preprocessing, model training, and validation techniques to improve classification performance.
- **Enhanced machine learning model performance** through iterative testing, validation, and optimization, increasing prediction accuracy and overall reliability.

Projects

Speech Emotion Recognition Using Machine Learning | *Python, FastAPI* **November 2025**

- **Built a speech emotion recognition system** to classify human emotions, including Happy, Sad, Angry, and Neutral, from recorded voice samples using machine learning techniques.
- **Constructed an end-to-end audio processing pipeline** incorporating feature extraction, data preprocessing, model training, and prediction to improve emotion classification accuracy.
- **Enhanced model performance** through validation, testing, and optimization on structured audio datasets, ensuring accurate emotion classification across diverse voice recordings.

ECG-Based Arrhythmia Detection Using a 1D Convolutional Neural Network

- **Established an ECG signal preprocessing pipeline** for physiological time-series data, enabling accurate arrhythmia classification through deep learning techniques.
- **Trained and validated a 1D Convolutional Neural Network (1D CNN)** for automated ECG arrhythmia detection using structured cardiac signal datasets, improving classification performance.
- **Prepared the trained deep learning model for edge deployment**, improving inference efficiency while reducing latency and memory usage on embedded devices. Option 2

Awards & Certifications

- **Deloitte Data Analytics Job Simulation (Forage, 2026):** Completed industry-focused training in business analytics, forensic technology, and data interpretation.
- **Infosys Springboard – Natural Language Processing in Practice (2025):** Acquired practical knowledge of NLP, text preprocessing, feature extraction, and Python-based language processing.

Education

B.Tech CSE (Artificial Intelligence and Machine Learning) **Graduated: 2026**
Dr. B.C. Roy Engineering College, Durgapur